

Question ID: 565428e1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{3n-1}$, where n is a constant, what is the value of n ?

Correct Answer: .5061, .5062, 41/81

Rationale

The correct answer is $\frac{41}{81}$. An expression of the form $\sqrt[n]{a^m}$, where m and n are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the left-hand side of the given equation is equivalent to $p^{\frac{2}{3}}$; thus, $p^{\frac{2}{3}} = t^{\frac{9}{7}}$. If $t = p^{3n-1}$, where n is a constant, then substituting p^{3n-1} for t in the equation yields $p^{\frac{2}{3}} = p^{\frac{9}{7}(3n-1)}$. It's given that $p > 1$; therefore, $\frac{2}{3} = \frac{9}{7}(3n-1)$. Multiplying each side of this equation by 21 yields $14 = 27(3n-1)$. Distributing multiplication over subtraction yields $14 = 81n - 27$. Adding 27 to each side of this equation yields $41 = 81n$. Dividing each side of this equation by 81 yields $\frac{41}{81} = n$. Therefore, if $t = p^{3n-1}$, where n is a constant, the value of n is $\frac{41}{81}$. Note that 41/81, .5061, .5062, and 0.506 are examples of ways to enter a correct answer.